K-ONT RESEARCH UPDATE · HEART-DISEASE VERIFICATION

Does K-Ont help machine learning more than OnT?

A controlled comparison of standard OnT and K-Ont (OnT extended to knowledge-graph individuals) as patient representations for heart-disease prediction, with four classic machine-learning models and a neural network.

Summary

+2.2	−0.5	−4.0	+4.3
AUC points, K-Ont vs OnT as the representation (mean of 5 models; 4 of 5 significant)	AUC points from adding K-Ont to the complete table (mean of 5 models)	AUC points from adding K-Ont with only 10% of the labels	AUC points from adding K-Ont when 3 columns are missing (5 of 5 significant)

- **K-Ont is a better representation than OnT.** It places every patient, including unseen ones, under the ontology's "patient" class (100% of unseen patients; OnT: disease status (294), heart disease (9)), keeps all 303 patients distinct (OnT: 76), and its vectors alone predict the diagnosis better than OnT's in every model.
- **With the complete table, no embedding improves the classifiers.** The knowledge graph holds the same 13 facts as the table, and this ontology adds no medical knowledge beyond them.
- **With few labels, the embeddings hurt.** Sixteen extra numbers per patient on 24 to 120 labelled patients make the classifiers overfit.
- **When columns are missing from the table, the knowledge-graph embedding brings their information back** and every model improves significantly; K-Ont edges OnT because OnT cannot hold numeric facts. An untrained encoder reading the same patient sentence does about as well, so the gain comes from embedding each patient as an individual, which K-Ont makes possible, rather than from K-Ont's extra training.

Shantharam P · K-Ont project · data: UCI Heart Disease (Cleveland, 303 patients) with the heart ontology and knowledge graph supplied by the guide · code: K-Ont repository, heart/

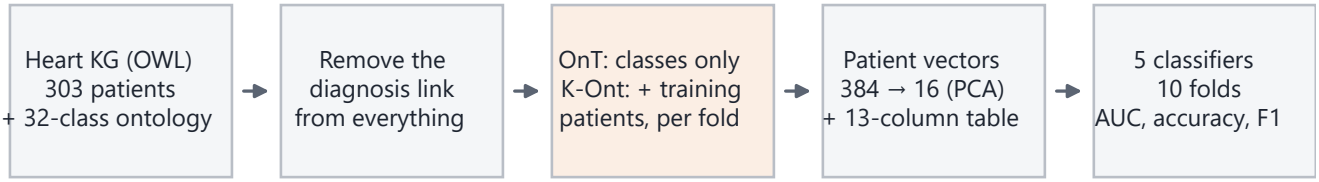
1. The question

Our battery-materials work showed that K-Ont, our extension of the Ontology Transformer (OnT) to knowledge-graph individuals, represents a knowledge graph better than standard OnT, which only embeds ontology classes. The guide asked whether this carries over to a practical task: if a machine-learning model is given ontology embeddings, does K-Ont improve its predictions more than OnT does? This report tests that on a second, independent domain: heart-disease diagnosis.

2. Data

- **Patients:** 303 patients from the UCI Cleveland heart-disease data, 13 clinical features (age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG, maximum heart rate, exercise-induced angina, ST depression, ST slope, number of major vessels, thallium scan) and a binary label (heart disease or not; 139 / 164).
- **Ontology:** 32 classes (e.g. typical angina ⊑ chest pain ⊑ abstract ⊑ entity) and 5 relations (has gender, has chest pain, has resting ECG result, has ST slope, has thallium scan result).
- **Knowledge graph:** the ontology plus the 303 patients as individuals, each with its 13 facts and a *hasDiagnosis* link. The graph lists patients in the same order as the table; ages were checked one by one. One ECG value, "ST wave anormality" (4 patients), has no class in the ontology and was mapped to its class "wave abnormality".

3. Method



Representations compared

- **Raw table** (baseline): the 13 features, categorical ones one-hot encoded, numeric ones standardised.
- **Standard OnT:** OnT trained on the ontology alone, with OnT's own DeepOnto data preparation (30 subsumption axioms), 20 epochs. OnT has no representation for an individual, so a patient is the concatenation of the OnT vectors of its five categorical values.
- **K-Ont:** OnT trained on the ontology plus the training patients as knowledge-graph individuals. Each patient is one sentence of its 13 facts (shown in section 4), trained as an instance of "patient" and linked by the five relations to its value concepts, with in-batch negatives; ontology rows are 15% of training; 3 epochs. A patient's representation is K-Ont's encoding of its sentence.
- **Plain encoder** (control): the untrained starting model (OnT-MiniLM-galen) reading the same patient sentence. It separates the value of K-Ont's training from the value of simply encoding the patient as text.

Leakage control

The diagnosis relation is removed from the ontology and never written into any sentence or training row. K-Ont is retrained inside every cross-validation fold on that fold's training patients only, so test patients are only ever encoded, never trained on. All vectors are mapped to the tangent space (logmap at the origin) and reduced to 16 numbers by PCA fitted on the training fold.

Classifiers and evaluation

Logistic regression, random forest (300–500 trees), RBF support-vector machine, histogram gradient boosting, and a multilayer perceptron (2 × 64 units, dropout 0.2, early stopping on an inner validation split, averaged over 3 seeds). Stratified 5-fold cross-validation repeated twice gives 10 folds, identical for every model and feature set. Metrics: ROC-AUC (main), accuracy and F1. Differences are paired fold by fold and tested with the Wilcoxon signed-rank test; "significant" means $p < 0.05$.

4. One patient, two ways

How each method turns the same knowledge-graph patient into numbers, and which other patients it considers most similar. Diagnoses are shown for reading only.

patient 001 · diagnosis no heart disease (never shown to the encoders) · seen by K-Ont in training (fold 0): yes

WHAT K-ONT READS: THE PATIENT AS A KNOWLEDGE-GRAPH INDIVIDUAL

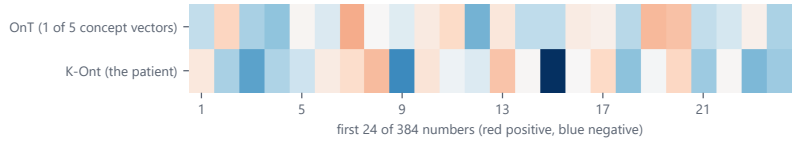
```
Type: patient
Gender: male
Chest pain: typical angina
Resting ECG: left ventricular
hypertrophy
ST slope: downsloping
Thallium scan: fixed defect
Age: 63
Resting blood pressure: 145 mm Hg
Serum cholesterol: 233 mg/dl
Fasting blood sugar over 120 mg/dl:
true
Maximum heart rate: 150
Exercise-induced angina: false
ST depression by exercise: 2.3
Major vessels coloured by
fluoroscopy: 0
```

WHAT STANDARD ONT CAN USE: FIVE ONTOLOGY CONCEPTS

male · typical angina · left ventricular hypertrophy · downsloping · fixed defect

OnT has no entry for a patient, so it embeds each concept separately (5 × 384 numbers). Age, blood pressure, cholesterol, heart rate, ST depression and vessels have no concept and are lost. K-Ont embeds the whole record as one point of 384 numbers.

THE VECTORS (FIRST 24 NUMBERS)



Nearest by OnT · 4 of 5 share the diagnosis

#	Patient	Diagnosis	Gender · chest pain · ECG · slope · thallium
1	patient 184	No heart disease	male · typical angina · left ventricular hypertrophy · downsloping · reseversable defect
2	patient 189	Heart disease	male · atypical angina · left ventricular hypertrophy · upsloping · reseversable defect
3	patient 289	No heart disease	male · atypical angina · left ventricular hypertrophy · upsloping · reseversable defect
4	patient 113	No heart disease	male · typical angina · left ventricular hypertrophy · flat · fixed defect
5	patient 060	No heart disease	male · typical angina · left ventricular hypertrophy · upsloping · normal

Nearest by K-Ont · 3 of 5 share the diagnosis

#	Patient	Diagnosis	Gender · chest pain · ECG · slope · thallium
1	patient 184	No heart disease	male · typical angina · left ventricular hypertrophy · downsloping · reseversable defect
2	patient 050	No heart disease	male · non anginal pain · left ventricular hypertrophy · downsloping · normal
3	patient 059	No heart disease	male · non anginal pain · left ventricular hypertrophy · downsloping · normal
4	patient 293	Heart disease	male · asymptomatic · normal · downsloping · fixed defect
5	patient 155	Heart disease	male · asymptomatic · left ventricular hypertrophy · downsloping · normal

patient 002 · diagnosis heart disease (never shown to the encoders) · seen by K-Ont in training (fold 0): yes

WHAT K-ONT READS: THE PATIENT AS A KNOWLEDGE-GRAPH INDIVIDUAL

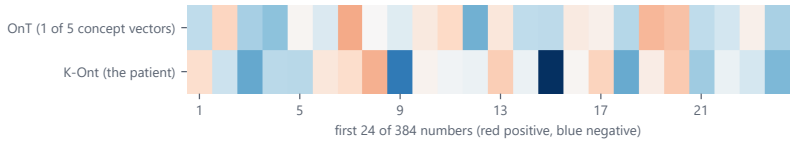
```
Type: patient
Gender: male
Chest pain: asymptomatic
Resting ECG: left ventricular
hypertrophy
ST slope: flat
Thallium scan: normal
Age: 67
Resting blood pressure: 160 mm Hg
Serum cholesterol: 286 mg/dl
Fasting blood sugar over 120 mg/dl:
false
Maximum heart rate: 108
Exercise-induced angina: true
ST depression by exercise: 1.5
Major vessels coloured by
fluoroscopy: 3
```

WHAT STANDARD ONT CAN USE: FIVE ONTOLOGY CONCEPTS

male · asymptomatic · left ventricular hypertrophy · flat · normal

OnT has no entry for a patient, so it embeds each concept separately (5 × 384 numbers). Age, blood pressure, cholesterol, heart rate, ST depression and vessels have no concept and are lost. K-Ont embeds the whole record as one point of 384 numbers.

THE VECTORS (FIRST 24 NUMBERS)



Nearest by OnT · 4 of 5 share the diagnosis

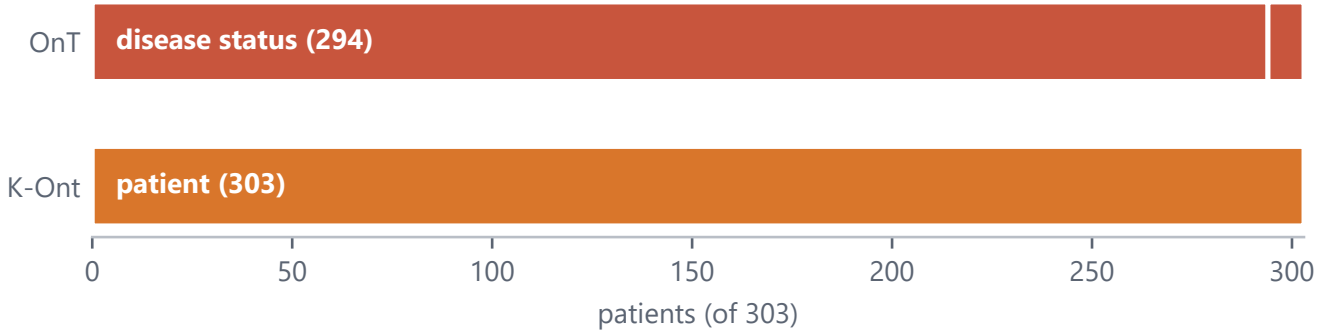
#	Patient	Diagnosis	Gender · chest pain · ECG · slope · thallium
1	patient 265	Heart disease	male · asymptomatic · left ventricular hypertrophy · flat · normal
2	patient 248	Heart disease	male · asymptomatic · left ventricular hypertrophy · flat · normal
3	patient 091	No heart disease	male · asymptomatic · left ventricular hypertrophy · flat · normal
4	patient 236	Heart disease	male · asymptomatic · left ventricular hypertrophy · flat · normal
5	patient 112	Heart disease	male · asymptomatic · left ventricular hypertrophy · flat · normal

Nearest by K-Ont · 5 of 5 share the diagnosis

#	Patient	Diagnosis	Gender · chest pain · ECG · slope · thallium
1	patient 196	Heart disease	male · asymptomatic · left ventricular hypertrophy · flat · normal
2	patient 265	Heart disease	male · asymptomatic · left ventricular hypertrophy · flat · normal
3	patient 248	Heart disease	male · asymptomatic · left ventricular hypertrophy · flat · normal
4	patient 236	Heart disease	male · asymptomatic · left ventricular hypertrophy · flat · normal
5	patient 112	Heart disease	male · asymptomatic · left ventricular hypertrophy · flat · normal

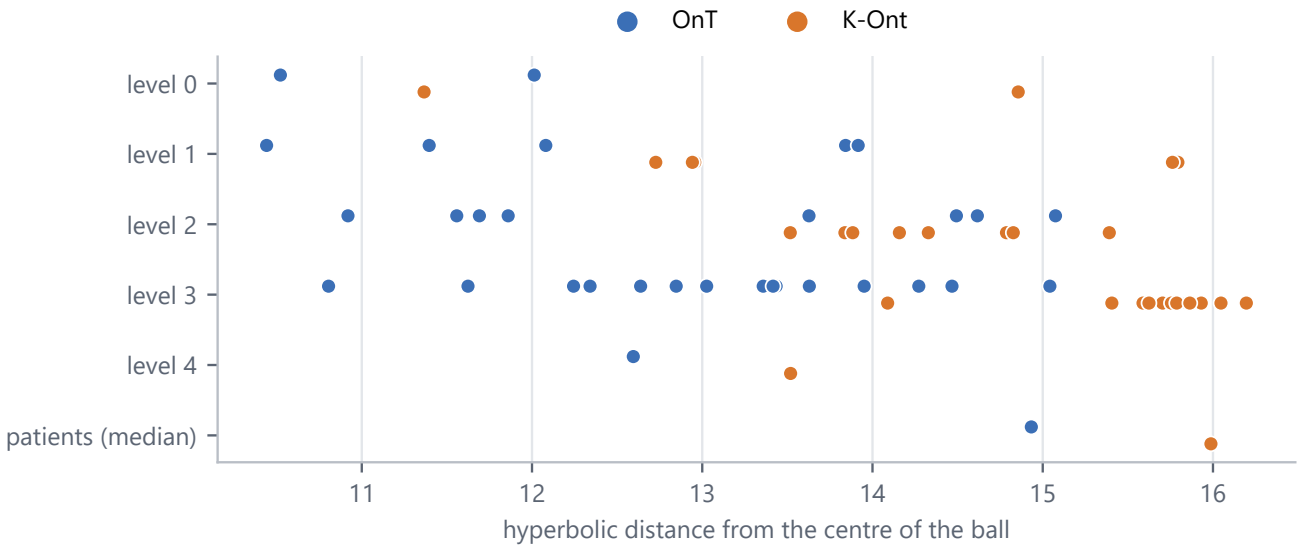
5. What the embeddings contain

Where each model places patients in the ontology



For every patient, the ontology class nearest to it by OnT's own hierarchy score. The correct answer is "patient". K-Ont places all patients, including the 61 test patients of fold 0 it never trained on, under "patient"; standard OnT, which was never shown an individual, places them near "disease status".

Hierarchy in hyperbolic space

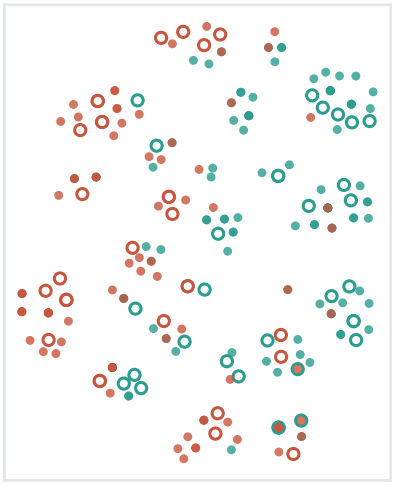


Distance from the centre of the Poincaré ball for every class, grouped by its level in the ontology (level 0 = top classes such as "entity"), and the median patient. In a well-formed hyperbolic embedding general classes sit near the centre and specific ones further out. K-Ont largely keeps this ordering and places patients, the most specific things, furthest out; OnT's classes are spread with little order.

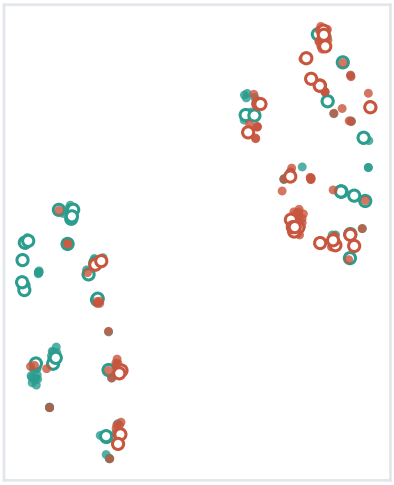
Map of all 303 patients

No heart disease Heart disease fold 0 test patient (unseen by K-Ont)

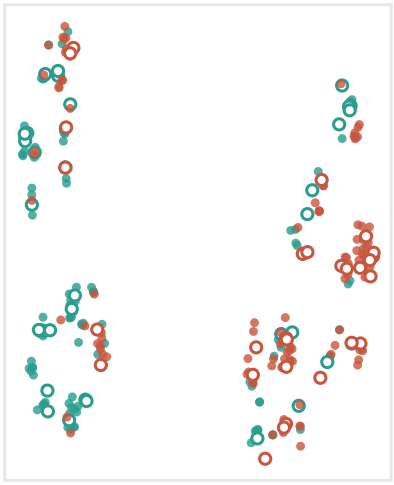
OnT (5 concepts)



K-Ont

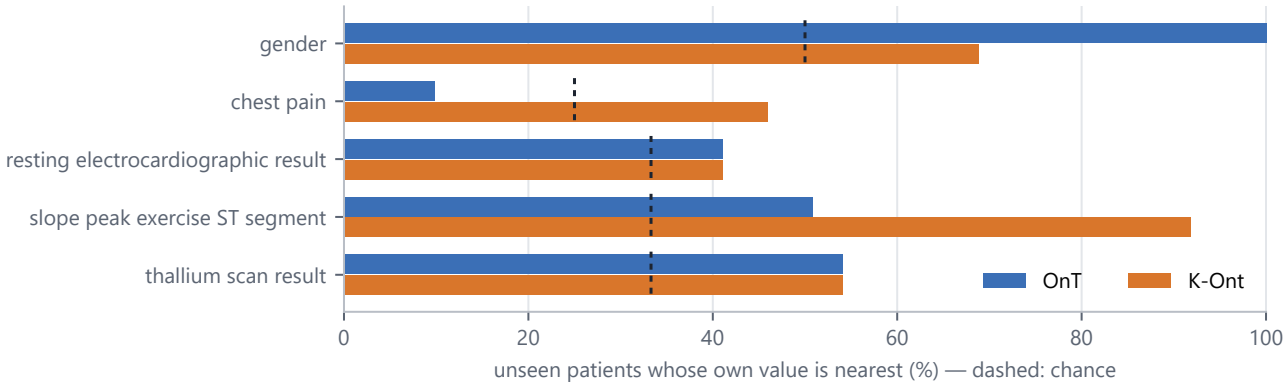


Plain encoder



Each representation reduced to two dimensions with t-SNE. OnT's five-concept representation gives many patients exactly the same point (76 distinct points for 303 patients), because patients with the same categories become identical. K-Ont and the plain encoder keep every patient distinct.

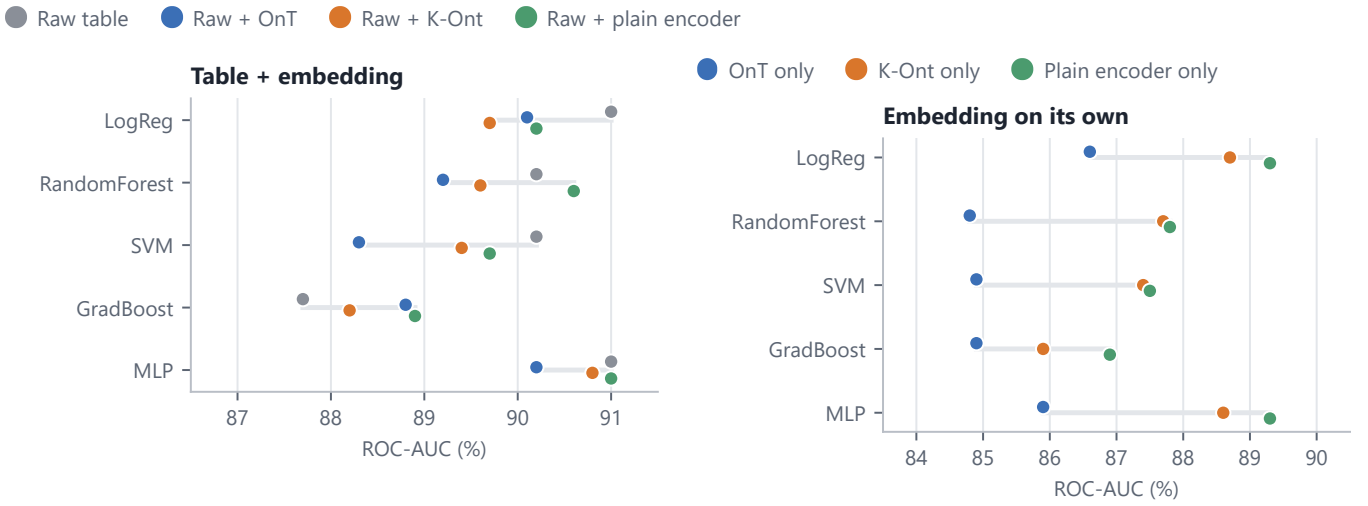
Checks on the representations



Relation check on fold 0's unseen patients: rotate every possible value concept by the relation (OnT's learned role rotation) and see whether the patient's own value is nearest. K-Ont beats chance on all five relations; OnT, whose relations were never trained on patients, is below chance on chest pain.

Check	OnT	K-Ont	Plain encoder	Chance
Unseen patients typed as "patient"	0.0%	100.0%	—	—
Relation: has gender	100.0%	68.8%	—	50.0%
Relation: has chest pain	9.8%	45.9%	—	25.0%
Relation: has resting electrocardiographic result	41.0%	41.0%	—	33.3%
Relation: has slope peak exercise ST segment	50.8%	91.8%	—	33.3%
Relation: has thallium scan result	54.1%	54.1%	—	33.3%
Same diagnosis among 10 nearest patients	73.5%	73.6%	72.2%	54.1%
Distinct patient points (of 303)	76	303	303	—

6. Test 1: prediction with the complete table



Mean ROC-AUC over 10 folds.

ROC-AUC (%)

Model	Raw table	Raw + OnT	Raw + K-Ont	Raw + plain encoder	OnT only	K-Ont only	Plain encoder only
LogReg	91.0	90.1	89.7	90.2	86.6	88.7	89.3
RandomForest	90.2	89.2	89.6	90.6	84.8	87.7	87.8
SVM	90.2	88.3	89.4	89.7	84.9	87.4	87.5
GradBoost	87.7	88.8	88.2	88.9	84.9	85.9	86.9
MLP	91.0	90.2	90.8	91.0	85.9	88.6	89.3

Accuracy (%)

Model	Raw table	Raw + OnT	Raw + K-Ont	Raw + plain encoder	OnT only	K-Ont only	Plain encoder only
LogReg	84.8	83.9	83.2	83.9	78.2	81.9	83.0
RandomForest	82.0	81.7	83.0	83.7	78.1	78.4	80.7
SVM	82.5	80.0	81.4	82.4	76.4	78.2	80.2
GradBoost	81.4	80.2	79.9	82.0	77.4	78.7	79.6
MLP	84.0	82.7	83.4	84.3	78.9	81.0	82.9

Paired differences in AUC (points; wins/losses over 10 folds; Wilcoxon p)

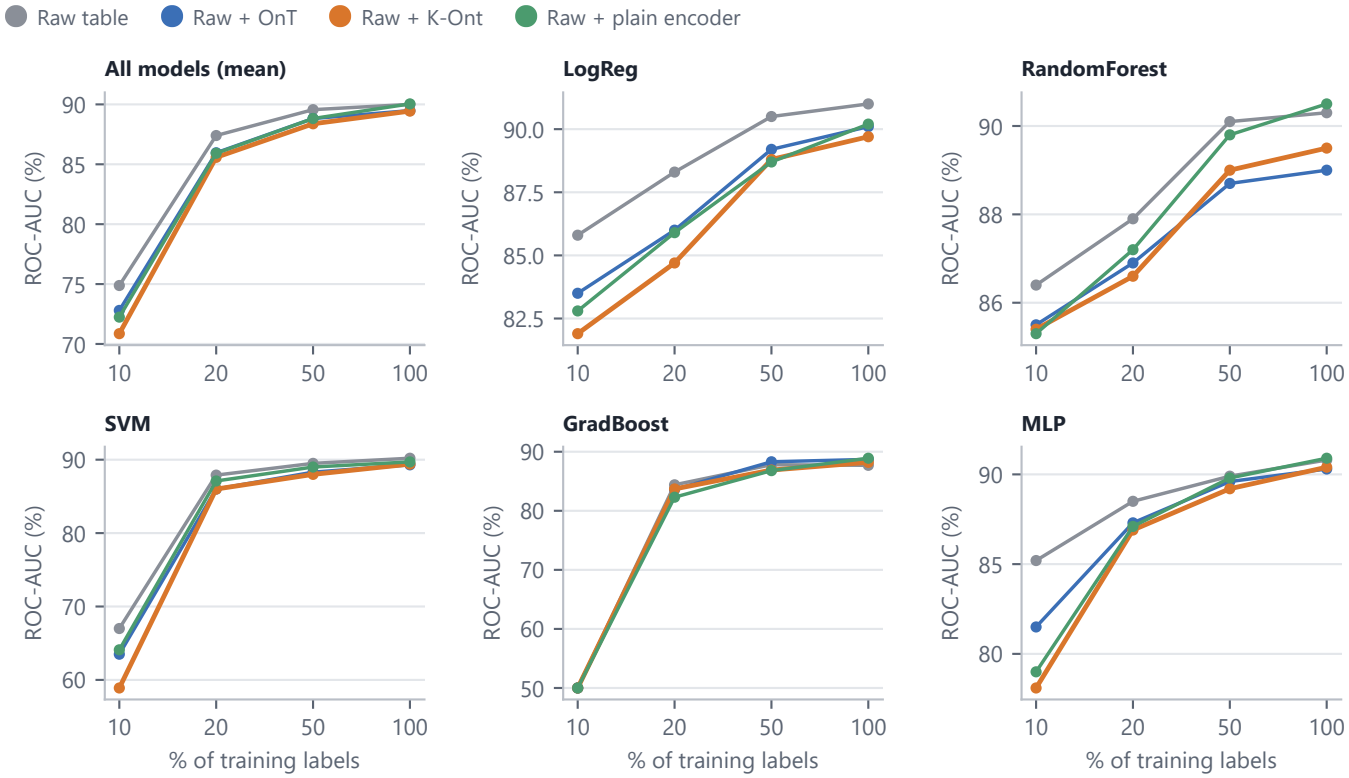
Green: significantly better; red: significantly worse.

Comparison	LogReg	RandomForest	SVM	GradBoost	MLP
K-Ont only vs OnT only	+2.1 (8/2, p 0.037)	+2.9 (9/1, p 0.014)	+2.5 (8/2, p 0.037)	+1.0 (7/3, p 0.322)	+2.6 (10/0, p 0.002)
Raw + K-Ont vs Raw + OnT	−0.4 (6/4, p 0.984)	+0.4 (7/3, p 0.232)	+1.0 (7/3, p 0.027)	−0.5 (4/6, p 0.625)	+0.6 (7/3, p 0.084)
Raw + K-Ont vs Raw table	−1.2 (2/7, p 0.203)	−0.6 (3/7, p 0.105)	−0.9 (2/8, p 0.105)	+0.5 (5/5, p 0.557)	−0.2 (5/5, p 0.826)
Raw + OnT vs Raw table	−0.9 (1/8, p 0.008)	−1.1 (2/8, p 0.012)	−1.9 (1/8, p 0.008)	+1.1 (8/2, p 0.105)	−0.8 (4/6, p 0.105)
Raw + K-Ont vs Raw + plain encoder	−0.5 (5/5, p 0.557)	−1.0 (1/9, p 0.029)	−0.4 (3/6, p 0.426)	−0.7 (2/8, p 0.232)	−0.2 (3/7, p 0.432)
K-Ont only vs Plain encoder only	−0.6 (4/6, p 0.336)	−0.1 (5/5, p 1.000)	±0.0 (6/4, p 0.822)	−1.0 (4/6, p 0.322)	−0.8 (4/6, p 0.375)

Reading. On its own, K-Ont's patient vector beats OnT's in every model (4 of 5 significantly). Added to the complete table, no embedding improves on the table alone, and OnT's concept vectors often make it significantly worse. K-Ont added to the table is close to the table alone and slightly ahead of OnT added to the table. The plain encoder on the same sentence performs like K-Ont.

7. Test 2: few labelled patients

The classifiers learn from 10, 20, 50 or 100% of the training patients' labels (stratified, 3 random subsets per fold). The embeddings, scalers and PCA use all training patients without labels, which is legitimate because no encoder ever sees a diagnosis.



Mean ROC-AUC by share of labels. Gradient boosting cannot split on 24 patients (minimum leaf size 20), so at 10% it predicts one value for everyone with every feature set.

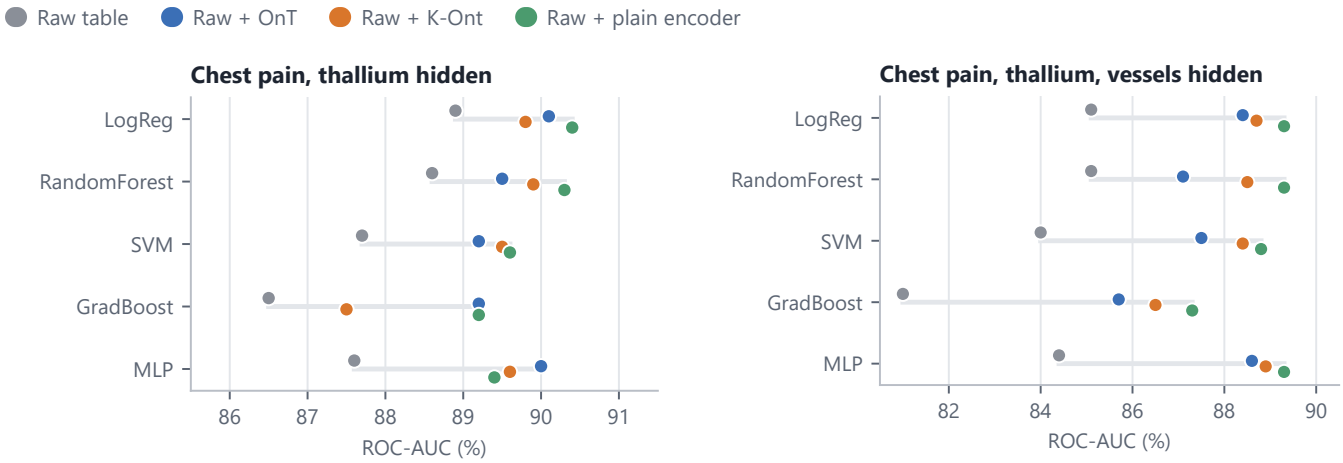
Raw + K-Ont vs the table alone (AUC points; Wilcoxon p over 30 runs, 10 at 100%)

Labels	LogReg	RandomForest	SVM	GradBoost	MLP
10%	**−3.9** (p <0.001)	**−1.0** (p 0.041)	−8.1 (p 0.241)	±0.0 (p 1.000)	**−7.1** (p 0.014)
20%	**−3.6** (p <0.001)	**−1.3** (p 0.004)	**−2.0** (p 0.002)	−0.7 (p 0.516)	**−1.6** (p 0.021)
50%	**−1.7** (p <0.001)	**−1.1** (p 0.003)	**−1.5** (p 0.008)	**−1.0** (p 0.018)	−0.7 (p 0.195)
100%	−1.2 (p 0.203)	**−0.9** (p 0.039)	−0.9 (p 0.105)	+0.5 (p 0.557)	−0.4 (p 0.492)

Reading. The hypothesis that pretrained embeddings help when labels are scarce does not hold here: with 10–50% of the labels, adding K-Ont lowers AUC, significantly in most models. The embeddings carry no information beyond the table, so the extra 16 numbers per patient only give the classifier more to overfit on when it sees 24 to 120 labelled patients.

8. Test 3: columns missing from the table

Chest pain type and thallium scan (and, in the second case, the number of major vessels) are removed from the table the classifiers see, while the knowledge-graph embeddings still contain them. OnT's concept vectors hold chest pain and thallium but cannot hold the number of vessels; K-Ont's patient sentence holds all three.



Hidden columns	Comparison	LogReg	RandomForest	SVM	GradBoost	MLP
chest pain, thallium	Raw + K-Ont vs Raw table	+0.9 (p 0.275)	+1.3 (p 0.232)	+1.8 (p 0.084)	+1.0 (p 0.275)	+2.0 (p 0.160)
chest pain, thallium	Raw + K-Ont vs Raw + OnT	−0.3 (p 0.846)	+0.4 (p 0.322)	+0.3 (p 0.375)	−1.7 (p 0.105)	−0.5 (p 0.322)
chest pain, thallium	Raw + K-Ont vs Raw + plain encoder	−0.6 (p 0.492)	−0.4 (p 0.359)	−0.1 (p 0.643)	−1.7 (p 0.105)	+0.2 (p 0.625)
chest pain, thallium, vessels	Raw + K-Ont vs Raw table	**+3.6** (p 0.010)	**+3.4** (p 0.012)	**+4.4** (p 0.004)	**+5.5** (p 0.002)	**+4.6** (p 0.002)
chest pain, thallium, vessels	Raw + K-Ont vs Raw + OnT	+0.4 (p 0.557)	**+1.4** (p 0.014)	+0.9 (p 0.082)	+0.8 (p 0.275)	+0.4 (p 0.557)
chest pain, thallium, vessels	Raw + K-Ont vs Raw + plain encoder	−0.5 (p 0.557)	−0.8 (p 0.084)	−0.4 (p 0.609)	−0.9 (p 0.432)	−0.3 (p 0.375)

Reading. This is where a knowledge-graph embedding pays off: with three columns hidden, adding K-Ont recovers +4.3 AUC points on average and improves all five models significantly (5 of 5). K-Ont is ahead of OnT, most clearly for the random forest, because OnT cannot represent the vessel count. The plain encoder on the same sentence recovers about as much, so the benefit comes from embedding each patient's facts as an individual.

9. Conclusions

- **K-Ont outperforms standard OnT as a representation of knowledge-graph individuals.** It types unseen patients correctly, keeps patients distinct, beats chance on all five relations, and its vectors predict the outcome better than OnT's. This agrees with our battery-materials results.
- **A knowledge-graph embedding helps a classifier when the graph holds information the classifier's input lacks** (test 3). It does not help when the input already contains everything (test 1), and it hurts with very few labels when it adds no new information (test 2).
- **K-Ont's training adds little beyond encoding the patient as text on this dataset.** The heart ontology is shallow and states no medical knowledge, so there is little for training to inject. The added value of K-Ont over OnT here is that it can embed individuals at all.

Limitations

- One small dataset (303 patients) and a shallow ontology; 10 folds give limited statistical power for differences of 1–2 AUC points.
- One K-Ont configuration (3 epochs, 16 PCA components); no tuning was done for either OnT or K-Ont, to keep the comparison fair.
- The relation check uses one fold's model; typing and maps use fold 0's K-Ont.

Recommended next steps

- Test on a knowledge graph with real domain knowledge that the table lacks: risk factors, drug–condition links, comorbidities, or links between patients.
- Report the missing-information setting (test 3) as the main use case of knowledge-graph embeddings, and keep the plain-encoder control in every comparison.
- For the battery project, the same lesson applies: K-Ont adds most where the knowledge graph carries structure or knowledge that the generator's prompt does not already contain.

Appendix: reproducibility

```
python heart/build_tbox.py --kg heart_disease_knowledge_graph.owl --out data/heart/heart_tbox.owl
python -m tbox.prepare --owl data/heart/heart_tbox.owl --out-dir data/heart/ont_tbox     # OnT's own data prep
python heart/kont_heart.py --stage prep | ont | folds | eval                            # test 1
python heart/report_data.py --stage fold0 | tests | visuals                             # tests 2-3, figures
python heart/report_datasets.py ; python heart/report_pdf.py                           # this report
```

Folds: sklearn RepeatedStratifiedKFold(n_splits=5, n_repeats=2, random_state=0). Base encoder: Hui97/OnT-MiniLM-L12-galen. Results: data/heart/results.json and data/heart/report/.